Paul Thielen

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Education

Carnegie Mellon University (CMU) - Pittsburgh PA *May 2027*

Master of Science in Electrical and Computer Engineering (AI/ML Concentration)

- Coursework: Computer Vision, Estimation Detection and Learning, Intro to Machine Learning

Technische Universiteit Eindhoven (TUE) – Eindhoven, NL *Jun 2024*

Bachelor of Science in Data Science

- GPA: **3.94/4.00** (Cum Laude)
- Minor: Cognitive Science & Artificial Intelligence

Internship Experience

Uber - Amsterdam, Netherlands *Jan 2023 – Jul 2023*

Data Scientist Intern

- Developed and implemented a **SARIMA model** in Python to project sales quota attainments, enabling the sales team to refine strategies and achieve targets
- Applied time series analysis and predictive modeling techniques in **Python** to identify high-impact sales strategies and improve decision-making, resulting in an **18%** increase in weekly sales income
- Wrote and automated **PrestoSQL** queries on large spatiotemporal datasets, reducing manual workload by **16 hours per week** with optimized query execution and report generation

Research Project Experience

Medical Data Under Shadow Attacks via Hybrid Model Inversion – AISTATS *Jun 2024 – May 2025*

- Developed MEDUSA, a hybrid gray-box **model inversion** framework combining gradient-based optimization with inverse modeling in **PyTorch**, in collaboration with a *Stanford AIMI* visiting researcher
- MEDUSA improved medical image reconstruction fidelity by up to **12%** over gradient-based reconstructions
- Demonstrated use of reconstructed medical images as synthetic training samples, attaining training performance within **5%** of real data benchmarks
- Invented and evaluated the KNN Smearing Defense method, reducing reconstruction quality by up to **64%**

Academic Project Experience

Food Insecurity in South Sudan - TUE Data Challenge *Feb 2024 – Apr 2024*

- Led a team project in collaboration with the *Zero Hunger Lab* to predict food insecurity risks in South Sudan using text data from news sources and climate data, earning a **9/10 grade**
- Trained **linear regression** models using climate data and conflict metric, and **GPT-3**-extracted mentions of violence from local and global news articles about South Sudan to forecast monthly risk scores with **Sklearn**
- Modeled impact of the Bentiu takeover using a **Bayesian Structural Time Series** model, identifying a **71%**statistically significant increase in food insecurity risk scores and a persistent post-conflict distribution shift

Predicting Crimes in Barnet, London - TUE Data Challenge *Apr 2023 – Jun 2023*

- Led a project as **Scrum Master** to implement OLS, Random Forest, SARIMA, and Prophet models to predict burglary trends in Barnet, training a model for each Lower Super Output Area (LSOA) to account for regional variations
- **Prophet** models achieved highest accuracy with an average R^2 of 0.83, capturing **83%** of burglary rate variability
- Implemented Gurobi **Integer Linear Programming** (ILP) solver to optimize officer allocation across LSOAs

Skills

Programming Languages:	Python, R, SQL, MATLAB
Deep Learning Frameworks:	PyTorch, TensorFlow, JAX, Keras, Hugging Face
Machine Learning & Data Science Tools:	Sklearn, Numpy, Pandas, Spark, Seaborn, OpenCV, PowerBI, Tableau
Cloud Computing Technologies:	Microsoft Azure, Amazon Web Services, Google Cloud Platform

Certificates

Machine Learning Track (DataCamp)	*Aug 2024*	Deep Learning Track (DataCamp)	*Jul 2024*
Microsoft Excel (Coursera)	*Jan 2025*	Microsoft AI & ML Engineering (Coursera)	*In progress*